



DATA SCIENCE

Project Problem Statement Submission Form

Sec No - 8	Team No - 15	Project Title - Crop Yield Prediction
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1. Abstract

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Crop yield prediction is an important application of Data Science that can help farmers and agricultural planners make better decisions regarding crop selection, resource utilization, and farm management. This project, “**Crop Yield Prediction Using Data Science and Machine Learning**,” aims to predict crop yield based on important agricultural factors such as soil quality, rainfall, temperature, humidity, fertilizer usage, irrigation, crop type, region, and previous yield.

The project follows the Data Science lifecycle, including data collection, preprocessing, data cleaning, exploratory data analysis (EDA), statistical analysis, visualization, model development, and evaluation. Python libraries such as **NumPy, Pandas, Matplotlib, and Seaborn** are used for data processing and visualization. Machine learning algorithms including **Linear Regression, Decision Tree, and Random Forest** are applied and compared using evaluation metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 score. The best-performing model is selected for final crop yield prediction.

A **Streamlit-based dashboard** is also developed to allow users to enter agricultural parameters and obtain an estimated crop yield. The system provides visual insights into the relationships between agricultural factors and crop productivity. The proposed solution demonstrates how Data Science and Machine Learning can be applied to agricultural data to support data-driven decision-making and improve the understanding of factors affecting crop yield.

2. Module Mapping

Module Name	Assigned Roll Number(s)	Signature
Data Collection & Preprocessing	2420030467 K.V.S Jaswant	
Data Analysis & Visualization	2420030737 Teja D	

3. Module Details

Module 1: Data Collection & Preprocessing – Roll No. 2420030467

Responsible for collecting the crop and agricultural dataset, checking data quality, handling missing or inconsistent values, identifying and handling noisy data and outliers, encoding categorical variables, and preparing the dataset for analysis and machine learning.

Module 2: Data Analysis & Visualization – Roll No. 2420030737

Responsible for performing exploratory data analysis and visualization. This includes analyzing the relationships between crop yield and factors such as rainfall, temperature, soil quality, fertilizer usage, humidity, crop type, and previous yield using statistical summaries, distribution plots, box plots, pairplots, correlation analysis, and heatmaps.

4. Frameworks / Platforms / Software Used

- **Python:** Primary programming language for data preprocessing, analysis, visualization, statistical analysis, and machine learning.
- **Pandas & NumPy:** Used for data cleaning, manipulation, numerical calculations, and dataset preparation.
- **Matplotlib & Seaborn:** Used for exploratory data analysis, charts, distribution plots, box plots, pairplots, and correlation heatmaps.
- **Scikit-learn:** Used for train-test splitting, preprocessing, machine learning model development, prediction, and evaluation.
- **Jupyter Notebook / Google Colab:** Used as the development environment for data analysis and machine learning experiments.
- **Streamlit:** Used to develop the interactive dashboard for crop yield prediction.

Professor: Sunandha

Guide Name and Signature (remarks if any) _____